Reviving Victorian Virtual Reality: A Toolkit for Restoring and Disseminating Historical Stereographs in Contemporary VR

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Stereo Photography and Challenges to its Dissemination

Stereo Photography was the first 3D immersive medium and a dominant form of optical entertainment in the 19th and 20th centuries. Invented in London shortly after the advent of commercially viable photography in 1839 (Gernsheim / Gernsheim 1969), stereography (sometimes called 'Victorian Virtual Reality') operates by encoding depth within the difference between two simultaneously captured side-by-side images (figure 1). When viewed in a stereoscope, the two images in a stereograph are fused into a single depth perspective by the viewer's brain, completely enveloping them in the captured scene. Due to its immersive nature, the medium was immensely popular in the Western world for nearly a century and significantly influenced the visual culture and entertainment media that succeeded it (Gurevitch 2013). During this period, stereography was used to document a variety of historical events and perspectives, including the First World War (Mendelson / Kitch 2011), global culture (DeLeskie 2000), and scientific subjects (Gavin 2018), among many others. Millions of stereographs from this period survive in cultural repositories around the globe, representing a treasure trove of priceless three-dimensional historical records of the past that have much to reveal not only in terms of history but also through diverse topics such as media archaeology and the nature of immersion.

Despite the wealth of digitized stereographs available online, disseminating them proves challenging due to the difficulty of transmitting the depth of information they encode. Traditional viewing through a stereoscope requires physical handling, which risks jeopardizing the fragile artifacts, the dominant form of digital dissemination as side-by-side stereo pairs fail to immerse the observer in their 3D historical contexts, and other popular 3D viewing methods such as anaglyphs are not as compelling as comparable visualization methods (Peixoto et al. 2019). However, Virtual reality, which has become increasingly ubiquitous in personal and museological spheres[1], offers avenues towards a renaissance for the stereographic medium.

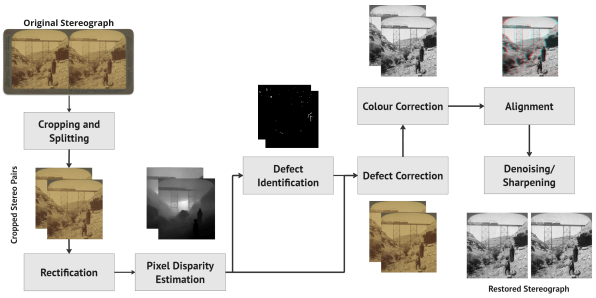

Figure 1: *"A wonder to the primitive inhabitants – Santa Fe train crossing Canon Diablo, Arizona" (1902), a stereograph whose full 3D relief is completely lost in this flat 2D archival scan.*

Yet, the visualization of this material on VR and stereoscopic displays is not as straightforward as simply scanning and displaying it (Luz / Peixoto 2015). Accumulated damage, exposure differences, noise, and misalignment generate asymmetries in the stereo view, triggering binocular rivalry and other sources of strain that our visual systems are highly sensitive to (Terzić / Hansard 2016).

Background & Related Work

Several projects have explored visualizing stereographs in VR and AR (Boehlert 2016; Taipina / Cardoso 2022; Sommerer et al. 1999; Sandheinrich / Hutson 2023); however, they all focus on creating demonstrators from small curated datasets, and none have thus far addressed systemic barriers to broader dissemination. Pipelines for creating digital stereograph datasets (Luo et al. 2020; Ricupero et al. 2019) still rely on manual processing and do not address sources of discomfort in VR visualization.

While automatic restoration methods have widely been explored for monoscopic photographs (Ardizzone et al. 2018; Wan et al. 2020), and a strong precedent exists for comfort optimization in 3D cinema (Baumgartner 2014; Mendiburu 2012), only one project has attempted the automatic restoration of stereographs to the best of our knowledge (Fanfani et al. 2021), but only addresses limited sources of discomfort, focusing more on visual improvement than comfort optimization.

Research into facilitating the dissemination of stereographs through VR is a largely untapped frontier for research, making the goals of our proposed toolkit novel in scope, breadth, and depth.

A Toolkit for Disseminating Digitized Historical Stereo Photographs

This project aims to build a set of open-source computational tools that archivists, restoration professionals, and stereo enthusiasts can use to process stereographs on a large scale and visualize them in VR. The kit also aims to explore new visualization techniques and augmentations, defining new visualization paradigms for the historical medium through virtual reality. The computational tools in our toolkit fall under three main categories: restoration, augmentation, and visualization.

Restoration

Our toolkit's core is an automatic end-to-end restoration pipeline that prepares digitized stereographs for VR. This involves cropping the two images out of their scanned frames and correcting various sources of discomfort. To this end, we identified eight restoration steps (Gowda-Storz / Kenderdine in press) that must be automated by examining manual stereo restoration and 3D cinema post-processing workflows employed by professionals. These eight steps, described in figure 2, process stereographs for both HMD and screen-based VR systems and are modular, with each step designed to be used in isolation. At its core, the pipeline is based on matching and comparing the two images based on disparity and using one image to correct the other. This, along with a focus on classical rather than generative methods, ensures transparency and minimizes the possibility of our algorithm introducing new, historically inaccurate content. The restoration pipeline is in active development and is shaped by input from the archives and restoration professionals who would benefit from it.

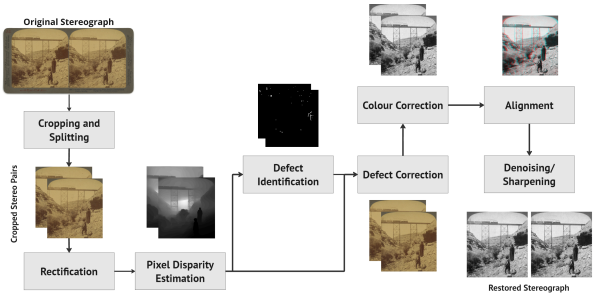

Figure 2: Overview of the eight sequential stages in our automated stereo-restoration pipeline, from initial frame extraction and disparity alignment through artifact correction and colour correction to the final restored stereograph.

Augmentation

In addition to preparing digitized stereographs for visualization in VR, our toolkit also explores the uncharted frontier of augmenting historical stereographs using computer vision methods to elevate them beyond their intended viewing format, leveraging the computing power and interactive modalities that contemporary VR devices offer. While this remains open-ended, we are working on reconstructing 3D scenes from the depth information encoded in stereo pairs, recolouring tools, automatic sonification tools, and NLP tools for linking the vast textual archives related to stereographs with the images they contextualize. As the augmentation tools deviate from the original material due to their use of generative AI, this facet of the toolkit is speculative. It aims to chart new paradigms for visualizing, interacting with, and creating narratives from stereoscopic archives that enhance and encourage their dissemination to the general public.

Visualization

The final facet of our toolkit is a set of visualization tools for deploying the restored and augmented stereographs on various VR systems, bridging the final gap between digitized stereo pairs and VR architectures. These visualization systems will be instrumental in evaluating the efficacy of the comfort improvements of our restoration pipeline and testing augmentations in a 3D context.

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Notes

1. Such as the advanced visualization systems at our laboratory: https://www.epfl.ch/labs/emplus/systems/

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